

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



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Order Instituting Rulemaking to
Enhance Demand Response in
California.

Rulemaking 25-09-004
(Filed September 18, 2025)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE
ADMINISTRATIVE LAW JUDGE'S RULING SEEKING COMMENTS ON DEMAND
RESPONSE STAFF PROPOSAL ON BRIDGE YEAR FUNDING AND MODIFYING
THE SCHEDULE FOR THE BRIDGE YEAR FUNDING ISSUE**

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Date: April 30, 2026

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments on the *Administrative Law Judge’s Ruling Seeking Comments on Demand Response Staff Proposal on Bridge Year Funding And Modifying The Schedule For The Bridge Year Funding Issue* (“Ruling”), issued by Administrative Law Judge (“ALJ”) Brandon T. Gerstle on March 10, 2026, and the *Administrative Law Judge’s Ruling Modifying Schedule on Bridge Year Funding Issue and Posing Supplemental Questions* issued by ALJ Brandon T. Gerstle on April 8, 2026, which extended the comment deadline to April 30, 2026.

I. INTRODUCTION.

VGIC is encouraged by the Commission Staff’s proposal to allow for two years of bridge funding for 2028 and 2029. Given that many of the matters discussed in this proceeding will materially impact demand response (“DR”) policy in California, it is prudent for the Commission to adopt bridge funding for near-term programs, instead of requiring a full DR application at this time.

VGIC largely supports the Staff proposal but recommends that the Commission also provide bridge funding for the Emergency Load Reduction Program (“ELRP”) Group A.5.

As highlighted by SCE in opening comments on the OIR, “Extending ELRP (or funding a similar new program) through the bridge period will provide continuity [...] avoiding the loss of critical megawatts (MW) as the state pursues its 7,000 MW load shift goal.”¹ ELRP provides critical DR capacity that no other program offers. Critically, Group A.5 is a unique participation pathway for electric vehicles (“EVs”), including bidirectional charging systems, and allows California to leverage grid value from EVs during grid emergencies.

II. RESPONSES TO RULING QUESTIONS.

4.1.a. Should the Commission authorize two years of bridge funding for PG&E, SCE, and SDG&E with programs and funding levels similar to the IOUs’ 2027 portfolio, excluding pilots, and including a budget to incorporate the SCT in the DR Cost Effectiveness Reporting tool?

Yes, the Commission should authorize two years of bridge funding for the investor-owned utilities (“IOUs”). However, VGIC believes that pilots should also be extended for two years and disagrees with the Staff proposal to exclude pilots from receiving funding. Specifically, ELRP should receive bridge funding for 2028 and 2029 to ensure emergency resources remain available while broader demand-response policies are developed.

4.1.b What is an appropriate level of funding for integrating the SCT into the 2030-2034 program cycle applications?

VGIC has no position on this topic at this time but may provide additional information in reply comments.

4.1.c Should the Commission consider other critical modifications to the IOUs’ 2027 DR programs and budgets if authorizing funding for the bridge years? If so, please recommend specific modifications to the programs and budget amounts for the two-year bridge year extension period.

¹ SCE Opening Comments on the Order Instituting Rulemaking at 5.

VGIC has no position on this topic at this time but may provide additional information in reply comments.

4.1.d Is there any other substantive feedback, not addressed above, that parties have on the Staff Proposal’s recommendation to authorize 2028-2029 bridge year funding?

VGIC has substantive feedback on extending ELRP for 2028-2029, which is currently not included in the Staff Proposal. This feedback is shared in VGIC’s response to question 4.1.h below.

4.1.e Should the Commission authorize an extension of time for the IOUs’ 2030-2034 applications until January 1, 2029?

VGIC believes that the latest the Commission should extend this deadline is January 1, 2029. In the last round of Demand Response Applications, utilities were delayed in filing, and the application was bifurcated into two phases. To ensure the proceeding can decide on 2030 programs before 2030, the Commission should provide at least a year to discuss the applications, and a timeframe closer to 18 months (e.g., applications due July 1, 2028) might be more appropriate. Otherwise, the Commission will likely need to approve more years of bridge funding or expedite consideration of a 2030 DR portfolio.

4.1.f Should the Commission exempt the bridge year portfolios from the SCT requirement?

Yes, given the timing of this proceeding, VGIC believes that an exemption from Societal Cost Test (“SCT”) requirements is reasonable to enable near-term bridge-funding authorization. Directing the IOUs to formulate SCT metrics, seeking stakeholder feedback, and conducting an in-depth Commission review of these metrics will take time. The ALJ Ruling proposes a schedule with a final decision on bridge funding before the end of 2026, which VGIC agrees is prudent. This accelerated timeline will make it difficult to allow for full consideration of the SCT. Instead, it is more appropriate for the utilities to evaluate the SCT in their next full-cycle DR applications.

4.1.g Is there any other procedural feedback, not addressed above, that parties have on the Staff Proposal's recommendation to authorize 2028-2029 bridge year funding?

VGIC has no position on this topic at this time but may provide additional information in reply comments.

4.1.h Should the ELRP pilot be extended as-is or with modifications? Should there be a sunset date?

Bridge funding should be approved for ELRP. At a minimum, ELRP Group A.5 should be extended for 2028-2029 to avoid a significant gap in California's demand response capabilities. ELRP Group A.5 provides a critical pathway for EV fleets and drivers to contribute emergency capacity to California's grid. Importantly, ELRP Group A.5 is the only utility program that enables both managed charging and bidirectional charging from EVs to support emergency grid needs. No other demand response or rate-based offering currently provides this functionality, and none are expected to be available after 2027. Absent bridge funding, ELRP will sunset, with no replacement program to enable many EVs, especially those that are part of bidirectional charging systems, to provide grid services.

VGIC does not believe ELRP will be a permanent mechanism for EVs to provide demand response and grid services. However, it is critical that this program remain in place while additional programs and/or rates are developed. VGIC looks forward to the development of additional programs and rates to drive load flexibility, but given the time required to fully develop those pathways within a new demand response application and/or rate applications, it is critical that ELRP be extended for 2028 and 2029. Otherwise, EV customers will be lost in the bridge period.

VGIC does recommend two minor modifications to ELRP to ensure that the program can function optimally in 2028 and 2029: (1) Align submetering pathway requirements for Group A.5 with Group A.4, and (2) Reserve ELRP funding for Group A.5 specifically to ensure sufficient

budget to support EV aggregations. These recommendations are further explained in our response to question 4.1.i below.

A. No alternative DR programs will be available to allow for bidirectional charging systems to support emergency reliability or other grid services.

There is no other utility-administered DR offering in California that allows bidirectional charging systems to participate. The Staff Proposal extends the following groups of programs, but none of these allow for bidirectional charging systems to participate:

- **Supply-side programs:** Because supply-side DR programs require participation in CAISO wholesale markets, exports are not compensated under these programs. The CAISO's Proxy Demand Response ("PDR") and Reliability Demand Response Resource ("RDRR") models do not recognize exports. Although the Distributed Energy Resource Aggregations ("DERA") CAISO model allows exports, it does not qualify for Resource Adequacy, has not been procured by California LSEs, and is not included in utility DR portfolios.
- **Load-modifying programs:** PG&E and SCE currently operate load-modifying programs, but they focus on large customers and do not recognize exports.
- **Emerging and enabling technology programs:** VGIC generally supports the utilities' emerging technology programs, which study new ways to leverage demand flexibility. The utilities have used some of this funding to study EV managed charging and load flexibility. However, these programs do not operate at a large scale and do not include bidirectional EV opportunities.

Additionally, beyond demand response, there are very few other offerings that can encourage bidirectional charging systems to support the grid. Currently, SCE does not have any rate offerings that encourage grid support from bidirectional charging systems. SCE does not compensate exports

from bidirectional charging systems in its dynamic rate pilot. SCE's proposed Vehicle-to-Grid Resource Proposal, which would have credited exports from bidirectional charging systems at a rate based on the Commission's adopted Avoided Cost Calculator, has been rejected by the Commission.² SDG&E has a limited dynamic export rate pilot for commercial customers, which will conclude at the end of 2027 and has seen zero customer enrollment.³ PG&E developed, and the Commission approved in 2022, a day-ahead hourly real-time pricing rate ("DAHRTTP") that would have credited exports from non-Net Energy Metering ("NEM") commercial EV customers, but PG&E never launched that rate, and it is now suspended.⁴ PG&E does credit bidirectional charging system exports in its Hourly Flex Pricing ("HFP") dynamic rate pilot, but only two of the 31 bidirectional charging customers incentivized under PG&E's Residential and Commercial Vehicle-to-Everything ("V2X") Pilots have elected to participate in the HFP rate pilot.⁵ This is likely due to the relative complexity of the rates involved, the lack of support from Automation Service Providers ("ASPs") who do not receive incentives to support customers (whereas non-V2X Pilot HFP ASPs do), and the high degree of uncertainty about the benefits that may accrue to the customer.

B. EV load reduction and bidirectional charging can provide significant value for California's reliability and ratepayers, but near-term programs are needed to ensure that maximum value is provided to ratepayers.

² *Proposed Decision Regarding Southern California Edison Company's Application to Establish Marginal Costs, Allocate Revenues and Design Rates* in Application ("A.") 24-03-019 at p.44-45.

³ SDG&E-01 in A.26-02-001.

⁴ See D.22-08-002 approving the final DAHRTTP rate, and *Executive Director Tesfai's letter of grant in response to Pacific Gas and Electric Company's Request for Extension of Time to Comply with D.22-08-002 DAHRTTP-BEV and GRC 2 RTP Pilots*, dated February 13, 2026, which has continued the suspension of the DAHRTTP rate.

⁵ Pacific Gas and Electric. *VGI Forum: VGI Pilot Use Cases*. March 25, 2026. https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/transportation-electrification/energization/vgi/2026-vgi-forum-combined-deck-all-slides_20260417.pdf. Slide 76.

California has already developed a substantial and consistent evidentiary record demonstrating that EV load reduction and bidirectional charging capability provide incremental and high-value grid services beyond charging scheduled around time-of-use (“TOU”) rate periods. Additionally, the Commission must, under SB 676, maximize the use of VGI.⁶ The Commission is already planning for approximately 13,500 MW of capacity from bidirectional charging systems by 2045 in its Integrated Resource Plan.⁷ PG&E’s Electrification Impact Study Part 2 found that 859 MW of additional demand flexibility could be unlocked through bidirectional EV exports.⁸ A recent CEC *Roadmap to Unlock the Benefits of Bidirectional Charging* estimates that bidirectional discharge can provide approximately three times the value of unidirectional managed charging.⁹ A wide variety of academic studies corroborate this conclusion: leveraging bidirectional charging systems in California materially increases available capacity and reduces system costs.¹⁰

⁶ SB 676 (Bradford, 2019), CA Pub Util Code § 740.16

⁷ California Public Utilities Commission (“CPUC”), Draft Inputs & Assumptions, 2024 – 2026 Integrated Resource Planning (IRP) at p.111. https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/integrated-resource-plan-and-long-term-procurement-plan-irp-ltpp/2024-2026-irp-cycle-events-and-materials/2025_draft_inputs_and_assumptions_doc_20250220.pdf

⁸ PG&E Electrification Impact Study Part 2 filed in R.21-06-017 at p.34.

⁹ CEC, *A Roadmap to Unlocking the Benefits of Bidirectional Charging* at p.3. <https://efiling.energy.ca.gov/GetDocument.aspx?tn=268952&DocumentContentId=106145>

¹⁰ See:

- Samantha Houston, David Reichmuth, and Mark Specht. *Harnessing the Power of Electric Vehicles: Vehicle-Grid Integration for a Cleaner, Cheaper, More Reliable California Electricity System*. June 2025. Union of Concerned Scientists. <https://doi.org/10.47923/2025.15888>. Accessed January 13, 2026.
- Sonia Martin, William A. Paxton, and Ram Rajagopal. *Residential vehicle-to-grid profits under dynamic pricing: The role of real-world charging behavior*. March 2026. Advances in Applied Energy. <https://www.sciencedirect.com/science/article/abs/pii/S0360544225024156?via%3Dihub>
- William Goldsmith et al., *The Utility Playbook: Turning EV Grid Risk into a \$30 Billion Opportunity*. 2025. EV.Energy and Brattle. <https://www.ev.energy/en-us/resources/value-of-managed-charging>
- Sunil Chhaya, PhD, et al. *Distribution System Constrained Vehicle-to-Grid Services for Improved Grid Stability and Reliability*. CEC. March 2019. <https://www.energy.ca.gov/sites/default/files/2021-06/CEC-500-2019-027.pdf>
- Jonathan Donadee et al. *Potential Benefits of Vehicle-to-Grid Technology in California: High Value for Capabilities Beyond One-Way Managed Charging*. Lawrence Livermore National Laboratory. IEEE Electrification Magazine. June 2019.

These systems can also be used today to help provide emergency capacity for the grid. The CEC’s Roadmap estimates that **18.5 GW of vehicle battery storage capacity is already available in California today, which is larger than the state’s stationary storage fleet.**¹¹ If the Commission can unlock additional bidirectional charging capabilities, EVs could become the state’s largest backup power reserve and change the way California plans for grid emergencies. One of the reasons bidirectional charging systems can provide so many grid benefits is their sheer size. The CEC’s recent analysis assumes the average light-duty EV has an 80 kWh battery and Level 2 bidirectional chargers can discharge at 11 kW.¹² EVs are an underutilized energy asset for both the grid and customers, and as EVs become more commonplace, customers will seek ways to get more from the system they have purchased.

In the absence of opportunities to participate in grid service programs, customers are likely to use these vehicles solely to meet on-site needs. The CEC’s Roadmap anticipates this outcome, noting that customers can prioritize vehicle-to-home (“V2H”) applications to offset their own peak TOU consumption.¹³ While such use cases can provide benefits, including an estimated 4.1 GW reduction in residential peak load by 2030, these systems would be limited in their broader system value if constrained only to passive TOU optimization. This trajectory risks pushing customers toward maximizing private benefits rather than aligning behavior with system needs,

<https://www.osti.gov/pages/biblio/1557041>

- Brian Tarroja and Eric Hittinger. *The value of consumer acceptance of controlled electric vehicle charging in a decarbonizing grid: The case of California*. UC Irvine. Energy (Volume 229). <https://www.sciencedirect.com/science/article/pii/S0360544221009397>
- Jessica Kerby, et al. *Assessing the Energy Equity Benefits of Mobile Energy Storage Solutions* 6th E-Mobility Power System Integration Symposium October 2022. https://www.pnnl.gov/sites/default/files/media/file/Assessing_the_energy_equity_benefits_of_mobile_energy_storage_solutions.pdf

¹¹ CEC, *A Roadmap to Unlocking the Benefits of Bidirectional Charging* at 2.

<https://efiling.energy.ca.gov/GetDocument.aspx?tn=268952&DocumentContentId=106145>

¹² Ibid.

¹³ The CEC refers to these systems as “grid disconnected”.

including emergency reliability needs. While non-exporting applications may coincide with grid benefits, customers acting independently will not necessarily discharge during the highest grid need or during emergency reliability events.

Technology development cycles are occurring now for the next generation of products that will be deployed over the next three to five years. Without clear policy direction, these systems may be optimized solely for customer-only use, including cycling for TOU arbitrage, rather than incorporating grid-supportive functions.

Extending ELRP Group A.5 both provides near-term capacity from EVs that California can use today and helps ensure that the managed charging and bidirectional charging industry continues to design systems that meet not only customer needs but also the broader needs of the grid. It is only through signals sent during times of need that these resources will be engaged to provide maximum value to ratepayers.

4.1.i If the ELRP pilot should be extended with modifications, should there be design changes made to it that align with the “Guiding Principles” staff proposal? Please explain any specific design changes recommended as part of the response.

VGIC recommends extending ELRP Group A.5 with only minor modifications. The Staff Proposal, at its core, is a straightforward extension of the 2027 budgets for DR programs. We see value in applying similar treatment to ELRP Group A.5.

While VGIC recommends minor program modifications below, it does not make sense to align ELRP changes specifically with the proposed Guiding Principles under consideration. Staff is not proposing to change any other DR programs to align with the proposed Guiding Principles. Aligning programs with new Guiding Principles will likely take more time than this track will provide. It is also premature to align any program with Guiding Principles that have not yet been

adopted. Notably, ELRP, especially Group A.5, already aligns with the core intent of the proposed guiding principles, but it is unique as an emergency program that does not provide Resource Adequacy. **Instead of trying to force large program changes at this stage, it is a better use of time and resources to extend ELRP Group A.5 with more minor modifications, and for the utilities and all stakeholders to consider how to align ELRP with *adopted* principles in the 2030-2034 DR applications.**

However, VGIC does recommend two minor modifications to ELRP for the 2028 and 2029 seasons:

- A. The Commission should remove the current requirement that ELRP Group A.5 participants participating via submetering pathways do so through the Commission’s adopted Plug-in EV Submetering Protocol (“PEV Submetering Protocol”). Instead, the submetering pathway for Group A.5 participants should align with the submetering requirements currently in place for ELRP Group A.4.**
- In the current Terms and Conditions, both Group A.4, for stationary storage devices, and Group A.5, for EV/VGI aggregations, allow for submetering. For Group A.5, the terms state “Upon adoption by the CPUC, the submeter must meet applicable standards established by the CPUC. Aggregators that elect to use sub-meter data for settlement purposes shall also comply with approved submetering services as outlined in the Aggregator Participation Agreement.”¹⁴ In contrast, Group A.4 only requires that aggregations “comply with approved submetering services as outlined in the

¹⁴ Pacific Gas & Electric Company Emergency Load Reduction Program (ELRP) Pilot Group A Terms and Conditions Pursuant to California Public Utilities Commission Decisions 21-03-056, 21-06-027, 21-12- 015, and 23-12-005, February 15, 2026, at p.16-17.

https://elrp.olivineinc.com/_files/pge/elrp/PGE-ELRP-Group-A-Terms-and-Conditions.pdf

Aggregator Participation Agreement.”¹⁵ Currently, the Commission has an approved PEV Submetering Protocol that requires adherence to specific standards. While VGIC appreciates that the CPUC has adopted a submetering protocol since ELRP was first established, that protocol was always intended to provide the accuracy and standards necessary to support sub-metered participation in *utility rates*, not necessarily for a program like ELRP. The initial PEV Submetering Pilot testing launched because compelling EV-specific TOU rates in the early and mid-2010s required customers to install a separate utility meter to participate, which was both cost-prohibitive and, in some cases, simply took up more physical space than certain residential customers could accommodate. The testing eventually resulted in the establishment of the PEV Submetering Protocol in 2022 to support customer participation in certain utility rates. The implementation advice letters following the 2022 adoption of the PEV Submetering Protocol detailed the specific utility rates in which customers could utilize the PEV Submetering Protocol instead of an entirely separate meter. The utilities’ current submetering webpages extensively and clearly detail the use of submetering for participating in *rates*, not for *DR pilots/programs*, further demonstrating the long-standing intent of all stakeholders, including the Commission, IOUs, and industry, to apply submetering to rate billing purposes, not DR pilots/programs.¹⁶

Meanwhile, other concluded and active *non-rate* DR pilots and programs that support EV participation, like SCE’s Charge Smart SoCal¹⁷ and PG&E’s evPulse

¹⁵ Ibid at p.16.

¹⁶ See <https://www.sce.com/factsheet/ev-charging-submeter-billing-option-information-sheet> and <https://www.pge.com/en/clean-energy/electric-vehicles/electric-vehicle-submetering-program.html#accordion-155817abb9-item-31ba5ee5b6>.

¹⁷ See more info at <https://www.sce.com/save-money/savings-programs/ways-to-save-at-home/what-is-demand-response>

Pilot¹⁸, correctly do *not* require devices to meet the PEV Submetering Protocol. Additionally, there are significant practical barriers to using the PEV Submetering Protocol today. One of these is the ineligibility of NEM/NBT customers, since VGIC estimates that 50-70% of California’s bidirectional charging early adopters have existing NEM or NBT systems. Another significant barrier is the requirement that the data be mediated by an approved Meter Data Management Agent (“MDMA”). It is currently difficult to become an MDMA and PG&E only has two certified MDMA’s¹⁹ and SCE only has four certified MDMA’s, none of which serve residential segments.²⁰ The requirement to use an MDMA to transfer data is not applicable to Group A.4. In order to enable comparable treatment between these two ELRP groups and not disproportionately disadvantage the relatively nascent EV/VGI aggregation market, the Commission should remove the requirement that Group A.5 customers that elect submetering do so through the PEV Submetering Protocol.

¹⁸See more info at: <https://www.weavegrid.com/news/weavegrid-launches-evpulse-for-northern-and-central-california-based-ev-drivers>

²⁰ See SCE's certified MDMA's at:

other grid services program, implementing an initial set-aside for these customers may be reasonable. Notably, in previous Commission treatment establishing and maintaining provisions for EV/VGI Aggregations, the CPUC has stated that VGI and bidirectional charging “is in its early stages of development.”²¹ The Commission has further acknowledged that EV/VGI aggregations are uniquely positioned within ELRP, stating that “the ELRP pilot [is] an opportunity to deploy and scale this [EV/VGI] resource.” VGI and bidirectional charging is unquestionably still in its early stages of development and, as such, the Commission’s findings are still as relevant as they were in previous Decisions and Resolutions establishing and preserving provisions for ELRP A.5. With this in mind, VGIC recommends that the Commission take a similar approach to the CEC’s recently updated guidance for DSGS Option 3:²² **a share of DSGS funding is specifically reserved for eligible EV customers, and remaining, unused funds are *then* provided to other customers.** By reserving specific, modest funding for Group A.5 for the 2028-2029 seasons, the Commission will ensure that these VGI resources are not left untapped and instead contribute to California’s grid needs.

4.1.j Are there lessons or elements from DSGS that could inform improving ELRP pilot design to benefit all ratepayers?

VGIC believes that some elements of DSGS could inform modifications to ELRP, including:

²¹ Resolution E-5267 at 15.

²² The CEC has reserved \$250,000 for EV aggregations for the 2026 season. Demand Side Grid Support (DSGS) Program Guidelines, Fifth Edition, <https://www.energy.ca.gov/publications/2026/demand-side-grid-support-dsgs-program-guidelines-fifth-edition>

- **Device-level metering:** As detailed above in VGIC’s response to question 4.1.i, DSGS Option 3 – as well as ELRP Group A.4 – leverages device-level measurement to facilitate participation, rather than the PEV Submetering Protocol, which has faced implementation challenges and limited eligibility.
- **Capacity payment structure:** DSGS Option 3 compensates participants based on a performance-based \$/kW-month capacity payment structure. This makes customer equipment investments, aggregator outreach to customers, and program administrator spending easier to predict and plan for.
- **Standardized aggregator payment terms:** DSGS Option 3 provides payments to aggregators, and it is up to those aggregators to create customer incentives compelling enough to encourage participation. This reduces the need to negotiate different contracts individually between IOUs and aggregators, lowering the administrative burden for both.

III. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the Ruling. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Executive Director

VEHICLE-GRID INTEGRATION COUNCIL

Date: April 30, 2026